



CLINICAL RESEARCH ARTICLE



Relationship between early maladaptive schemas (EMSs), adverse childhood experiences and mental health in adulthood among transgender individuals

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ABSTRACT

Background: Transgender individuals experience a high level of distress over the lifespan and are at a higher risk of experiencing mental health conditions.

Objective: The present study mainly aimed to investigate the complex relationship between early maladaptive schemas (EMSs), adverse childhood experiences and mental health conditions in adulthood among transgender individuals.

Method: 220 participants completed an anonymous online study assessing sociodemographic and medical status, experience of childhood adversity (CTQ), EMSs (YSQ-S3), transgender specific quality of life (ETLI), current depression (PHQ-8), and generalised anxiety severity (GAD-7).

Results: Transgender individuals tended to have a high prevalence for experiencing childhood adversity as well as for depressive and generalised anxiety symptoms and low quality of life in adulthood, while assigned female at birth (AFAB) transgender tend to be more vulnerable than assigned male at birth (AMAB) transgender. In general, transgender individuals showed an elevated pronounced level of EMSs, while AFAB transgender tended to focus more on the negative aspects in life, to strive more to pursue perfection and tended more to be highly critical towards others and especially themselves, as well as tended to believe that people should be harshly punished for making mistakes than AMAB transgender. According to the network analysis, schema domain and their EMSs were related to all assessed psychopathological outcomes, especially the schema domains *Disconnection and Rejection* and *Impaired Autonomy and Performance* seemed to be related to experience of childhood adversity and quality of life in adulthood.

Conclusions: Transgender individuals seem to experience a high level of childhood adversity, which possibly leads to the development of an elevated level of EMSs. EMSs and their schema domains are associated with elevated depressive and generalised symptoms, as well as lower quality of life.

La relación entre esquemas desadaptativos tempranos (EDTs), experiencias adversas en la infancia y salud mental en la adultez entre personas transgénero

Antecedentes: Las personas transgénero experimentan un nivel alto de malestar emocional a lo largo de su vida y tienen mayor riesgo de sufrir problemas de salud mental.

Objetivo: El presente estudio tuvo como objetivo principal investigar la compleja relación entre esquemas desadaptativos tempranos (EDTs), experiencias adversas en la infancia y condiciones de salud mental en la adultez entre personas transgénero.

Método: 220 participantes respondieron un estudio anónimo en línea para evaluar el estado sociodemográfico y médico, experiencia de adversidades en la infancia (CTQ), EDTs (YSQ-S3), calidad de vida específica de las personas transgénero (ETLI), depresión actual (PHQ-8) y gravedad de la ansiedad generalizada (GAD-7).

Resultados: Las personas transgénero tendían a presentar una alta prevalencia de experiencias adversas en la infancia, así como también síntomas depresivos y de ansiedad generalizada y baja calidad de vida en la adultez, por otro lado, las personas transgénero asignadas como mujeres al nacimiento (AFAB en sus siglas en inglés) tienden a ser más vulnerables que los transgéneros asignados como varones al nacimiento (AMAB en sus siglas en inglés). En general, los individuos transgénero mostraron un nivel elevado y pronunciado de EDTs, mientras que las personas transgénero AFAB tendían a focalizarse más en los aspectos negativos de la vida, a esforzarse más por alcanzar la perfección y a ser más críticos con los

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HIGHLIGHTS

- Transgender individuals tend to have a high prevalence for childhood adversity as well as depressive and generalised anxiety symptoms, and low quality of life in adulthood.
- Assigned female at birth (AFAB) transgender individuals seem to be more vulnerable than assigned male at birth (AMAB) transgender.
- Adverse childhood experiences emerged as having a central impact on the presentation of early maladaptive schemas in adulthood, which interfere with mental health conditions in adulthood among transgender individuals.

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demás y especialmente consigo mismos, además tendían a creer que las personas deberían ser castigadas duramente por cometer errores en comparación con las personas transgénero AMAB. De acuerdo con el análisis de redes, el dominio de esquema y sus EDTs se relacionaban con todos los resultados psicopatológicos evaluados, especialmente los dominios de esquema Desconexión y Rechazo y Deterioro de la Autonomía y Rendimiento que parecían estar relacionados con experiencias adversas en la infancia y calidad de vida en la adultez.

Conclusiones: Los individuos transgénero parecen experimentar un nivel alto de adversidades en la infancia, lo que posiblemente conduce al desarrollo de un nivel elevado de EDTs. Los EDTs y sus dominios de esquema se asocian con síntomas elevados de depresión y ansiedad generalizada, así como también una menor calidad de vida.

1. Introduction

Transgender is an umbrella term used to describe people whose gender identity or gender expression differs from their sex assigned at birth (Coleman et al., 2012). Transgender individuals experience a high level of distress related to their gender incongruence (APA, 2015). Although identifying as transgender itself is not considered a mental health disorder, it is associated with vulnerability to a range of mental health conditions. For example, experiencing childhood gender non-conformity is associated with numerous childhood psychosocial stressors including rejecting parenting with parental detachment and lack of warmth (Mohta et al., 2017), discrimination (Bradford et al., 2013) and victimisation (Giovanardi et al., 2018). Gender non-conformity is an indicator of increases risk of childhood abuse (Roberts et al., 2012). A growing number of research highlights the disproportionate burden of adverse childhood experiences on transgender individuals (Charak et al., 2023; Engel et al., 2023; Friedman et al., 2011; Higgins et al., 2025). In a study by Biedermann et al. (2021) over 30% of transgender individuals reported severe to extreme childhood adversities and only 7% showed no childhood adversities. Transgender individuals are at increased risk of experiencing adverse childhood experiences, such as childhood abuse and neglect, compared to cisgender counterparts (Schnarrs et al., 2019; Thoma et al., 2021; Tobin & Delaney, 2019).

The higher prevalence of childhood adversities and mental health conditions is often associated with gender-diverse identities. It is crucial to consider the impact of minority stress (Meyer, 2003). An extended gender minority stress framework by Hendricks and Testa (2012) posits an interplay between gender identity and expression as distinctive constructs with sociocultural expectations. Transgender individuals have due to the incongruence of gender identity and expression with the sociocultural expectations an increased risk of experiencing minority stressors (Testa et al., 2015). By adopting this perspective, the focus is shifting from inherent characteristics of transgender identities to the social and psychological

stressors that arise from societal prejudice and discrimination.

Transgender individuals had a high prevalence of experiencing potentially traumatic events with 98% experienced at least one potentially traumatic event, relative to 56% of cisgender individuals from the general population, and 91% reported multiple events. Among those, 42% experienced at least one event that was transgender bias-related and 17.8% developed symptoms of PTSD (Shipherd et al., 2011). In a survey by the European Union Agency for Fundamental Rights (FRA, 2012) with 6,579 self-identified transgender individuals in the European Union every second transgender individual reported at least one event of violence or harassment.

Transgender individuals as gender minorities have a high prevalence of depressive symptoms (52%) and anxiety symptoms (38%), which is significantly higher compared with non-gender minorities (Reisner et al., 2016). Among LGBT people, transgender individuals were at greatest risk with a lifetime prevalence of 46.7% for non-suicidal self-harm (Liu et al., 2019). Across 42 studies, 55% of transgender individuals ideated about and 29% attempted suicide in their life span with AMAB transgender ideating more often about suicide while AFAB transgender more often attempted suicide (Adams et al., 2017). Rimes et al. (2019) found that AFAB transgender were more likely to report mental health conditions and current psychopathology, as well as childhood sexual abuse than AMAB transgender, with both reporting a higher level of psychopathology than the general population. Although the current study results in regard to the differences between AFAB and AMAB transgender are inconsistent, there is a clear indication that both, AFAB as well as AMAB transgender, are heavily burdened. In terms of psychopathology, the experienced quality of life (QoL) of transgender individuals seems to be a crucial aspect (Tagay et al., 2018). Studies investigating quality of life (QoL), a salient indicator for wellbeing of an individual, in transgender individuals demonstrated varying results. The majority of studies have found lower QoL in transgender individuals compared to the general population

(Lindqvist et al., 2017). A systematic review by Nobili et al. (2018) concluded that transgender individuals reported poorer QoL compared to the general population, regardless of the investigated domain. Regarding QoL, no differences were found between AFAB and AMAB transgender individuals (Auer et al., 2017).

Young et al. (2003) proposed that specific Early Maladaptive Schemas (EMSs) develop during childhood as adaptive responses to one's aversive environment, form a template that guides the interpretation of later experience and may elicit significant impairment (Arntz & Jacob, 2017). EMSs are theorised to be a result of unmet core emotional needs in childhood, primarily due to the interaction between adverse childhood experiences and emotional temperament (Young et al., 2003). EMSs are defined as dysfunctional and pervasive patterns of memories, thoughts, and physical sensations regarding the self, relationships, and world that develop in childhood and adolescent years and elaborate throughout the life (Young et al., 2003, p. 7). Young et al. (2003) identified 18 EMSs with five schema domains congruous with the unmet childhood needs: *Disconnection and Rejection*, *Impaired Autonomy and Performance*, *Impaired Limits*, *Other-Directedness*, and *Overvigilance and Inhibition*. The current findings suggested that EMSs are over the time stable character traits rather than unsteady and simple reflection of symptomatology (Wang et al., 2010). The current research suggested that EMSs that develop due to adverse childhood experiences mediate the relationship between adverse childhood experiences and later symptoms (Lumley & Harkness, 2007). In a systematic review with 33 studies the experience of childhood adversity, emotional and physical abuse, as well as emotional, physical and sexual abuse, was small linked with EMSs in adulthood, with the strongest link to a history of childhood emotional abuse (Pilkington, Bishop, et al., 2022). EMSs are considered as important vulnerability factors for the development and maintenance of a range of psychopathologies. EMSs were found to be related to depression (e.g. Bishop et al., 2022), anxiety disorders (e.g. Tariq et al., 2021), PTSD (e.g. Price, 2007), and intimate partner violence victimisation (e.g. Pilkington, Noonan, et al., 2021), suicidal ideation and self-harm (e.g. Pilkington, Younan, et al., 2021) and far more. Considering the adverse childhood experiences that transgender individuals experience, it is expected that EMSs among transgender individuals are elevated. A limited number of studies indicated that transgender individuals seem to feel especially disconnected and rejected by society with AMAB transgender tend to be more vulnerable (Hatami & Ayvazi, 2013; Simon et al., 2011). Schema therapy is an integrative approach that has proven to enhance the well-being of individuals by identifying and modifying of EMSs

through a combination of cognitive, behavioural, and experimental techniques (Young et al., 2003).

1.1. Research aims of the present study

The current literature suggests a complex relationship between childhood adversities, EMSs and psychopathology (Bishop et al., 2022). To date, there is still a research gap in exploring how schema domains and their EMSs are related to the psychopathology of transgender individuals. First, the present study aimed to assess the EMSs and psychopathology among transgender individuals. Second, it sought to investigate differences between AFAB and AMAB transgender. Third and mainly, it aimed to analyse how schema domains and its EMSs are related to the different psychopathological outcomes. It is of interest whether the schema domains and their EMSs are associated in a reliable way with adverse childhood experiences and mental health conditions in adulthood to provide insights into the impact of adverse childhood experience on the presentation of schema domains and their impact on mental health conditions in adulthood. This has the potential to develop targeted interventions in the health care for transgender individuals.

2. Materials & methods

2.1. Participants, procedure and study design

Over the time course of 13 weeks, 29 October 2021–28 January 2022, 220 individuals, consisting of 107 AFAB transgender and 113 AMAB transgender individuals, took part in an online cross-sectional study. The participants were recruited via transgender support groups and counselling centres, psychotherapy and expert practices, clinics and personal contacts. The eligibility requirement was identifying as transgender, adult age (≥ 18 years), sufficient command of the German language and internet access for all participants. Electric informed consent was given and confirmed by the start of the survey. The study participation was anonymous, voluntary, and could be terminated at any time without negative consequences for the participant. See Table 1 for sociodemographic characteristics and Table 2 for medical and psychological status of the participants.

2.2. Measures

The online study contained items on sociodemographic, medical and psychological information, as well as validated assessment instruments were used to assess adverse childhood experiences and early maladaptive schemas, generalised anxiety, depression symptoms and transgender specific quality of life. All used

Table 1. Sociodemographics.

	Total N = 220 (%)		
	AFAB ^a N = 107 (%)	AMAB ^b N = 113 (%)	
Non-binary gender identity			
Binary	175 (79.5)	80 (74.8)	95 (84.1)
Non-binary	45 (20.5)	27 (25.2)	18 (15.9)
Age [Years]			
M (SD)	36.59 (14.73)	29.39 (10.49)	43.41 (14.95)
Marital status			
Single	82 (37.3)	44 (41.1)	38 (33.6)
In a relationship	61 (27.7)	39 (36.4)	22 (19.5)
Married	53 (24.1)	18 (16.8)	35 (31.0)
Married living separatly	7 (3.2)	1 (0.9)	6 (5.3)
divorced	8 (3.6)	1 (0.9)	7 (6.2)
Widowed	3 (1.4)	0 (0.0)	3 (2.7)
Other	6 (2.7)	4 (3.7)	2 (1.8)
Living situation			
Living alone	80 (36.4)	35 (32.7)	45 (39.8)
Living with partner	58 (26.4)	30 (28.0)	28 (24.8)
Living alone with	4 (1.8)	1 (0.9)	3 (2.7)
child(ren)			
Living with partner and	21 (9.5)	6 (5.6)	15 (13.3)
child(ren)			
Living with parents	28 (12.7)	15 (14.0)	13 (11.5)
In an institution	2 (0.9)	1 (0.9)	1 (0.9)
Shared apartment	22 (10.0)	15 (14.0)	7 (6.2)
Other	5 (2.3)	4 (3.7)	1 (0.9)
Child(ren)			
Yes	48 (21.8)	8 (7.5)	40 (35.4)
No	172 (78.2)	99 (92.5)	73 (64.6)
Education level			
Completed university	69 (31.4)	32 (29.9)	37 (32.7)
degree			
High school degree	89 (40.5)	46 (43.0)	43 (38.1)
('Fach-)Abitur')			
Secondary school degree	36 (16.4)	17 (15.9)	19 (16.8)
('Realschule')			
Secondary school degree	17 (7.7)	6 (5.6)	11 (9.7)
('Hauptschule/ Volksschule')			
Special school degree	1 (0.5)	0 (0.0)	1 (0.9)
('Förderschule')			
School prematurely	1 (0.5)	1 (0.9)	0 (0.0)
cancelled			
Currently pupil	5 (2.3)	3 (2.8)	2 (1.8)
Other	2 (0.9)	2 (1.9)	0 (0.0)
Professional status			
Employed	98 (44.5)	39 (36.4)	59 (52.2)
Self-employed	19 (8.6)	8 (7.5)	11 (9.7)
Not employed	16 (7.3)	6 (5.6)	10 (8.8)
In training	69 (31.4)	48 (44.9)	21 (18.6)
Retired	18 (8.2)	6 (5.6)	12 (10.6)

Note: ^aassigned female at birth.^bassigned male at birth.

assessment instruments have been found to have satisfactory internal consistency (Hinz et al., 2017; Klinitzke et al., 2012; Kriston et al., 2013; Kroenke et al., 2009; Tagay et al., 2018), but have not been validated for use with transgender individuals:

Childhood Trauma Questionnaire (CTQ; German version; Klinitzke et al., 2012; Wingenfeld et al., 2010): The CTQ assesses the exposure to adverse childhood experiences. The questionnaire contains six subscales with 28 items on a five-point Likert scale (1 = 'never true' to 5 'very often true'). The five subscales that were all considered in the present study were (1.) *emotional abuse*, (2.) *physical abuse*, (3.) *sexual abuse*, (4.) *emotional neglect*, and (5.) *physical neglect*. Severity scores can be calculated, ranging

Table 2. Medical and psychological information.

	Total N = 220 (%)		
	AFAB ^a N = 107 (%)	AMAB ^b N = 113 (%)	
Age of realising being transgender [Years]			
M (SD)	16.63 (11.05)	14.57 (7.87)	18.58 (13.12)
Coming-out as transgender			
Yes	209 (95.0)	106 (99.1)	103 (91.2)
No	11 (5.0)	1 (0.9)	10 (8.8)
Time of coming-out as transgender [Years] ^c			
M (SD)	6.43 (6.84)	6.23 (6.04)	6.61 (7.54)
Living out being transgender			
Completely in daily life	133 (60.5)	68 (63.6)	65 (57.5)
With few exceptions	35 (15.9)	19 (17.8)	16 (14.2)
Restricted to close social environment	40 (18.2)	16 (15.0)	24 (21.2)
Just for themselves	8 (3.6)	2 (1.9)	6 (5.3)
Solely online	2 (0.9)	1 (0.9)	1 (0.9)
Not at all	2 (0.9)	1 (0.9)	1 (0.9)
Current status of gender reassignment process			
Not intended	73 (33.2)	44 (41.1)	29 (25.7)
Not yet but intended	35 (15.9)	14 (13.1)	21 (18.6)
Solely optical changes	22 (10.0)	15 (14.0)	7 (6.2)
Hormone Substitution	137 (62.3)	72 (67.3)	65 (57.5)
Sex reassignment surgery specially planned	31 (14.1)	14 (13.1)	17 (15.0)
After sex reassignment surgery	51 (23.2)	21 (19.6)	30 (26.5)
Psychotherapeutic support	89 (40.5)	47 (43.9)	42 (37.2)
Time of sex reassignment surgery [Years]			
M (SD)	1.59 (4.42)	1.55 (4.32)	1.62 (4.53)
Medication			
Any medication	58 (26.4)	22 (20.6)	36 (31.9)
Estradiol	69 (31.4)	1 (0.9)	68 (60.2)
Testosterone	81 (36.8)	79 (73.8)	2 (1.8)
Anti-androgens	23 (10.5)	0 (0.0)	23 (20.4)
GnRH-agonists	3 (1.4)	2 (1.9)	1 (0.9)
Progesterones	19 (8.6)	4 (3.7)	15 (13.3)
Medical diagnosis ^d			
Any medical diagnosis	49 (22.3)	19 (17.8)	30 (26.5)
Genderdysphoria	117 (53.2)	59 (55.1)	58 (51.3)
Depression	82 (37.3)	50 (46.7)	32 (28.3)
Anxiety disorder	34 (15.5)	20 (18.7)	14 (12.4)
Anxiety disorder with depression	15 (6.8)	7 (6.5)	8 (7.1)
PTSD	16 (7.3)	12 (11.2)	4 (3.5)
Borderline PD	12 (5.5)	8 (7.5)	4 (3.5)
Other PD	13 (5.9)	7 (6.5)	6 (5.3)
Autism Spectrum Disorder	14 (6.4)	8 (7.5)	6 (5.3)
Other	37 (16.8)	26 (24.3)	11 (9.7)

Note: ^aassigned female at birth.^bassigned male at birth.^ctime in years since the individual came out as transgender.^dThe medical diagnoses are self-reported and were assessed by a multiple-choice response option with the opportunity to add not-listed diagnoses. The not-listed diagnoses are not presented in the table, as they were only minimally represented.

from 'none-minimal', 'minimal-moderate', 'moderate-severe', to 'severe-extreme'. It is important to emphasise that this measure only captures a specific range of adverse childhood experiences.

Young Schema-Questionnaire – Short Form 3 (YSQ-S3; German version; Kriston et al., 2013): The YSQ-S3 consists of 90 items assessing 18 early maladaptive schemas (EMS), categorised into five domains. The YSQ utilises a six-point Likert scale (1 = 'strongly disagree' to 6 = 'strongly agree') with higher values

indicate a stronger presence of the respective schema. The five domains are (1.) *Disconnection and Rejection*, (2.) *Impaired Autonomy and Performance*, (3.) *Impaired Limits*, (4.) *Other-Directedness*, and (5.) *Over-vigilance and Inhibition*.

Essen Transidentity Quality of Life-Inventory (ETLI; Tagay et al., 2018): The ETLI consists of 30 items assessing the specific aspects of quality of life (QoL) in transgender individuals. The ETLI utilises a 4-points Likert Scale (0 = 'not at all' to 3 = 'very much') with higher values indicate a higher QOL. The ETLI assesses a global transgender specific QoL and the dimensions (1.) *physical QoL*, (2.) *mental QoL*, (3.) *social QoL*, and (4.) *QoL through disclosure of the gender dysphoria*. For each subscale a specific cut-off point was identified, which indicates a sufficient quality of life.

Patient Health Questionnaire-8 (PHQ-8; Kroenke et al., 2009): The PHQ-8 consists of eight items assessing self-reported depression in the past two weeks on a 4-point Likert scale (0 = 'never experiencing the symptom' to 3 = 'experiencing the symptom nearly every day'). Scores of ≥ 5 , ≥ 10 , ≥ 15 , and ≥ 20 points represent cut-off points for mild, moderate, intermediate, and severe depression symptomatology, respectively.

Generalised Anxiety Disorder Scale-7 (GAD-7; Spitzer et al., 2006): The GAD-7 consists of seven items assessing self-reported anxiety and its severity over the past two weeks on a 4-point Likert scale (0 = never to 3 = almost every day). Scores of ≥ 5 , ≥ 10 , and ≥ 15 represent cut-off points for mild, moderate, and severe anxiety, respectively.

2.3. Data analysis

Data analyses were carried out using the *Statistical Programme for Social Sciences SPSS* version 29 (IBM Corporation, 2023 New York) and *Rstudio* (R Core Team, 2023).

First, descriptive analyses were conducted to determine the characteristics of examined variables related to mental health, psychopathology and EMSs. Second, in the presence of non-normal distributions, as indicated by preliminary analyses, Mann-Whitney-U tests were calculated to investigate differences between AFAB transgender and AMAB transgender. The level of significance for all analyses was set to $\alpha = 0.05$.

Last, a network analysis in *Rstudio* was performed to investigate the relationship between schema domains and its EMSs (YSQ-S3) and adverse childhood experiences (CTQ), generalised anxiety (GAD-7), depression symptoms (PHQ-8) and transgender specific quality of life (ETLI) in the present transgender population, using the packages *qgraph*, *igraph*, *bootnet*, and *EGAnet* (Epskamp et al., 2012; Epskamp et al., 2018; Golino & Epskamp, 2017). The network

was estimated and visualised, the centrality indices were computed, and exploratory graph analyses (EGA; Golino & Epskamp, 2017) were performed. Furthermore, to assess the network's stability of the centrality indices and accuracy of the edge weights using *bootnet*, the bootstrapping procedures were conducted. As nodes were the following measures selected: neglect and abuse as adverse childhood experiences, the five schema domains according to Young et al. (2003), generalised anxiety, depressive-ness, and global transgender specific quality of life. A partial correlation network of 10 nodes with five groups resulted. For each group, a Gaussian graphical model (GGM; Epskamp et al., 2018) was estimated. Via the graphical LASSO (Least Absolute Shrinkage and Selection Operator; Friedman et al., 2008) algorithm, the GGMs were regularised. The model selection was based on the extended Bayesian information criterion (EBIC; Chen & Chen, 2008) with a tuning parameter of .5. *Degree centrality* and three centrality indices *strength*, *closeness*, and *betweenness* were estimated. Edge weight variation, significance of edge weight and node strength differences, and correlation stability of the centrality indices were conducted via bootstrap procedures.

3. Results

3.1. Prevalence and differences

The prevalence of depressive and generalised anxiety symptoms, transgender specific quality of life, and childhood adversity are shown in Table 3. Mann-Whitney-U-Tests were calculated to determine differences in mental health and psychopathology between AFAB and AMAB transgender, see Table 4. Regarding EMSs and schema domains, Mann-Whitney-U-Tests were performed to examine differences between AFAB and AMAB transgender, see Table 5.

A network analysis was conducted to analyse the relationship between schema domains according to Young et al. (2003), adverse childhood experiences and mental health conditions in adulthood. In total, 17 out of 100 possible edges emerged. Among them, 13 edges were positive (76.5%) and four (23.5%) were negative. A graphical representation of the network with displayed edge weights is shown in Figure 1.

4. Discussion

The current study examined the mental health of transgender individuals and how EMSs and their schema domains are related to adverse childhood experiences and mental health conditions in adulthood. Transgender individuals tended to have a high prevalence for experiencing childhood adversity as well as for depressive and generalised anxiety

Table 3. Prevalence of depressive and generalised anxiety symptoms, transgender specific quality of life, and childhood adversity, stratified by gender identity.

	Total N = 220 (%)	
	AFAB ^a N = 107 (%)	AMAB ^b N = 113 (%)
<i>PHQ-8 – Depressive symptoms</i>		
None	56 (25.5)	23 (21.5)
Mild	64 (29.1)	31 (29.0)
Moderate	61 (27.7)	29 (27.1)
Moderately severe	31 (14.1)	20 (18.7)
Severe	8 (3.6)	4 (3.7)
<i>GAD-7 – Generalised Anxiety</i>		
None	76 (34.5)	27 (25.2)
Mild	73 (33.2)	38 (35.5)
Moderate	48 (21.8)	27 (25.2)
Severe	23 (10.5)	15 (14.0)
<i>ETLI – Quality of Life</i>		
<i>Physical QoL</i>		
Low	68 (30.9)	34 (31.8)
Sufficient	152 (69.1)	79 (69.9)
<i>Mental QoL</i>		
Low	114 (51.8)	71 (66.4)
Sufficient	106 (48.2)	48 (42.5)
<i>Social QoL</i>		
Low	114 (51.8)	55 (51.4)
Sufficient	106 (48.2)	52 (48.6)
<i>QoL through openness</i>		
Low	68 (30.9)	41 (38.3)
Sufficient	152 (69.1)	66 (61.7)
<i>Global QoL</i>		
Low	119 (54.1)	65 (60.7)
Sufficient	101 (45.9)	42 (39.3)
<i>CTQ – Childhood adversity</i>		
<i>Emotional abuse</i>		
None-minimal	93 (42.3)	35 (32.7)
Minimal-moderate	45 (20.5)	22 (20.6)
Moderate-severe	28 (12.7)	18 (16.8)
Severe-extreme	54 (24.5)	32 (29.9)
<i>Physical abuse</i>		
None-minimal	175 (79.5)	81 (75.7)
Minimal-moderate	11 (5.0)	6 (5.6)
Moderate-severe	14 (6.4)	7 (6.5)
Severe-extreme	20 (9.1)	13 (12.1)
<i>Sexual abuse</i>		
None-minimal	149 (67.7)	63 (58.9)
Minimal-moderate	21 (9.5)	13 (12.1)
Moderate-severe	28 (12.7)	17 (15.9)
Severe-extreme	22 (10.0)	14 (13.1)
<i>Emotional neglect</i>		
None-minimal	61 (27.7)	28 (26.2)
Minimal-moderate	69 (31.4)	32 (29.9)
Moderate-severe	36 (16.4)	17 (15.9)
Severe-extreme	54 (24.5)	30 (28.0)
<i>Physical neglect</i>		
None-minimal	110 (50.0)	53 (49.5)
Minimal-moderate	51 (23.2)	24 (22.4)
Moderate-severe	33 (15.0)	15 (14.0)
Severe-extreme	26 (11.8)	11 (9.7)

Note: ^aassigned female at birth.^bassigned male at birth.

symptoms and low quality of life in adulthood. Moreover, they showed an elevated pronounced level of EMSs, with AFAB transgender tended to focus more on the negative aspects in life, to strive more to pursue perfection and tended more to be highly critical towards others and especially themselves, as well as tended to believe that people should be harshly punished for making mistakes than AMAB transgender. According to the network analysis, especially the schema domains *Disconnection and Rejection* and *Impaired Autonomy and Performance* seemed to be

related to experience of childhood adversity and quality of life in adulthood.

4.1. High prevalence for adverse childhood experiences and mental health conditions in adulthood

45.4% of the participants reported at least a moderate level of depression severity and 32.3% at least a moderate level of generalised anxiety severity. This indicates that almost half of the participants and a third of the participants experience current depressive and generalised anxiety symptoms, respectively. On average, more than half of the participants reported a low global quality of life. Further, over a third of the participants reported at least a moderate to severe exposure level to emotional abuse (37.2%) and emotional neglect (40.9%), as well as an almost quarter of the participants expressed at least a moderate to severe exposure level to sexual abuse (22.7%) and physical neglect (26.8%), and almost a fifth of the participants a moderate to severe exposure level to physical abuse (15.5%). These results are in accordance with recent literature indicating that transgender individuals have a high prevalence for depressive and anxiety symptoms (Reisner et al., 2016) and experiencing a high level of childhood adversity (Biedermann et al., 2021; Schnarrs et al., 2019) and a low quality of life (Lindqvist et al., 2017). In comparison, AFAB transgender showed a significantly higher level of anxiety severity and exposure level for emotional and sexual abuse, while AMAB transgender showed a significantly elevated level of quality of life through openness.

4.2. Elevated EMSs and differences among AFAB and AMAB transgender

Regarding EMSs, on average all schema domains and nearly all EMSs were rated at almost 'slightly more true than untrue'. According to these data, it can infer that almost all EMSs seem to be present among transgender individuals. This also accords with the high prevalence for experiencing childhood adversity in the present study, as adverse childhood experiences are considered pivotal in the development of EMSs (Young et al., 2003). The presence of almost all EMSs can be considered as an important vulnerability factor among transgender individuals, as the current research indicates that EMSs are related to a manifold range of psychopathology (e.g. Nicol et al., 2020). In comparison, AFAB transgender showed significant higher scores regarding the schema domain *Overvigilance and Performance* with the EMSs *Negativity/Pessimism*, *Unrelenting Standards/Hypercriticalness* and *Punitiveness* than AMAB transgender. This finding indicates that AFAB transgender seem more likely to suppress one's spontaneous feelings and

Table 4. EMSs differences between AFAB and AMAB transgender individuals.

	α^c	Total M (SD)	AFAB ^a M (SD)	AMAB ^b M (SD)	Mann-Whitney-U-Test		
					U	Z	p
Disconnection and Rejection	0.80	2.99 (1.04)	3.00 (1.09)	2.98 (0.99)	5994.00	−0.11	.913
Abandonment / Instability	0.89	3.21 (1.45)	3.27 (1.54)	3.16 (1.38)	5728.50	−0.67	.501
Mistrust / Abuse	0.86	2.91 (1.25)	2.93 (1.31)	2.89 (1.19)	5999.00	−0.10	.921
Emotional Deprivation	0.94	2.63 (1.54)	2.53 (1.47)	2.73 (1.61)	5682.50	−0.78	.438
Defectiveness / Shame	0.90	2.60 (1.37)	2.62 (1.44)	2.58 (1.30)	5977.00	−0.15	.884
Social Isolation / Alienation	0.90	3.60 (1.37)	3.63 (1.38)	3.57 (1.37)	5890.50	−0.33	.742
Impaired Autonomy and Performance	0.77	2.26 (0.92)	2.31 (0.93)	2.11 (0.91)	5655.50	−0.83	.408
Dependence / Incompetence	0.86	2.07 (1.06)	2.17 (1.04)	1.97 (1.08)	5178.50	−1.85	.064
Vulnerability to Harm or Illness	0.83	2.34 (1.21)	2.39 (1.26)	2.30 (1.17)	5931.00	−0.24	.807
Enmeshment / Undeveloped Self	0.81	2.07 (1.09)	2.03 (1.07)	2.10 (1.11)	5905.50	−0.30	.765
Failure	0.93	2.55 (1.41)	2.64 (1.47)	2.47 (1.35)	5698.00	−0.74	.459
Impaired Limits	0.43	2.77 (0.93)	2.78 (0.92)	2.76 (0.94)	5994.00	−0.11	.913
Entitlement / Grandiosity	0.77	2.44 (1.04)	2.38 (0.95)	2.50 (1.11)	5835.00	−0.45	.655
Insufficient Self-Control / Self-Discipline	0.88	3.10 (1.27)	3.18 (1.32)	3.02 (1.23)	5650.00	−0.84	.401
Other-Directedness	0.61	3.05 (0.88)	3.13 (0.94)	2.98 (0.83)	5540.00	−1.07	.284
Subjugation	0.89	2.68 (1.30)	2.77 (1.33)	2.58 (1.26)	5565.00	−1.02	.308
Self-Sacrifice	0.84	3.66 (1.15)	3.77 (1.19)	3.55 (1.11)	5366.00	−1.44	.149
Approval-Seeking / Recognition-Seeking	0.79	2.83 (1.09)	2.84 (1.09)	2.81 (1.09)	6012.00	−0.07	.943
Overvigilance and Inhibition	0.75	3.23 (0.98)	3.41 (0.98)	3.06 (0.95)	4828.00	−2.58	.010*
Negativity / Pessimism	0.88	3.23 (1.36)	3.42 (1.45)	3.05 (1.26)	5122.00	−1.96	.050*
Emotional Inhibition	0.89	2.91 (1.37)	2.99 (1.33)	2.83 (1.41)	5598.50	−0.95	.343
Unrelenting Standards / Hypercriticalness	0.86	3.61 (1.30)	3.86 (1.29)	3.37 (1.26)	4709.00	−2.84	.005*
Punitiveness	0.79	3.18 (1.15)	3.39 (1.23)	2.99 (1.04)	4959.50	−2.31	.021*

Note: ^aassigned female at birth.^bassigned male at birth.^cCronbach's alpha.

*Indicates a significant difference <0.05.

impulses, as well to meet rigid, internalised rules and expectations about performance and ethical behaviour than AMAB transgender. These results suggest that AFAB transgender seem to tend more on focusing on the negative aspects in life like suffering and adversity of life, strive more to pursue perfection and tend more to be highly critical towards others and especially themselves, as well as tend to believe that people should be harshly punished for making mistakes. This finding is contrary to previous studies which have suggested that AMAB transgender tend to feel more socially isolated and vulnerable than AFAB transgender (Hatami & Ayvazi, 2013; Simon et al., 2011). It seems possible that these results are

due to the sample composition. The previous studies investigated EMSs among transgender individuals in a clinical context, whereas the present study examined EMSs among transgender individuals in a nonclinical online context. It is possible that the results in the present study were influenced by the lack of verification of the diagnostic status. Otherwise, the present study considers a more representative number of transgender individuals than the previous studies of Hatami and Ayvazi (2013) and Simon et al. (2011). These findings raise intriguing questions regarding the nature of differences between AFAB and AMAB transgender considering mental health, psychopathology and EMSs. Further work is required to ascertain

Table 5. Differences between AFAB and AMAB transgender individuals on depressive and generalised anxiety symptoms, transgender specific quality of life, and childhood adversity.

	α^c	Total M (SD)	AFAB ^a M (SD)	AMAB ^b M (SD)	Mann-Whitney-U-Test		
					U	Z	p
PHQ-8	0.88	8.95 (5.62)	9.66 (5.72)	8.27 (5.46)	5207.00	−1.78	.075
GAD-7	0.89	7.22 (5.15)	8.25 (5.06)	6.24 (5.06)	4651.00	−2.96	.003*
ETLI							
Physical QoL	0.82	1.47 (0.67)	1.37 (0.65)	1.56 (0.68)	5153.50	−1.90	.058
Mental QoL	0.89	1.94 (0.58)	1.89 (0.59)	1.98 (0.56)	5575.00	−1.00	.318
Social QoL	0.82	1.65 (0.80)	1.68 (0.80)	1.62 (0.81)	5866.00	−0.38	.703
QoL through openness	0.83	2.02 (0.74)	1.90 (0.65)	2.14 (0.81)	4502.00	−3.29	.001*
Global QoL	0.78	1.81 (0.54)	1.75 (0.53)	1.86 (0.55)	5302.50	−1.58	.115
CTQ	0.82						
Emotional abuse	0.91	11.38 (5.67)	12.70 (5.87)	10.12 (5.21)	4409.00	−3.48	<.001*
Physical abuse	0.87	6.82 (3.45)	7.16 (4.10)	6.50 (2.68)	5914.00	−0.31	.754
Sexual abuse	0.93	6.88 (3.88)	7.46 (4.32)	6.34 (3.33)	4970.50	−2.75	.006*
Emotional neglect	0.90	13.36 (5.25)	13.66 (5.49)	13.08 (5.02)	5685.00	−0.77	.444
Physical neglect	0.63	8.19 (3.23)	8.34 (3.56)	8.05 (2.90)	6024.00	−0.05	.963

Note: ^aassigned female at birth.^bassigned male at birth.^cCronbach's alpha.

*Indicates a significant difference <0.05.

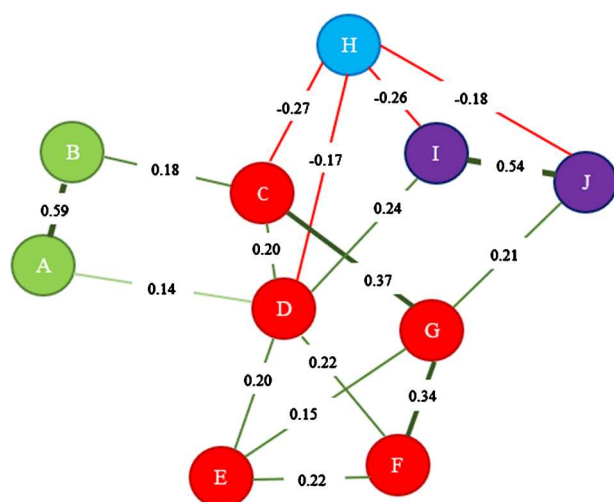


Figure 1. Visualised partial correlation network. A = experience of childhood abuse, B = experience of childhood neglect, C = schema domain *Disconnection and Rejection*, D = schema domain *Impaired Autonomy and Performance*, E = schema domain *Impaired Limits*, F = schema domain *Other-Directedness*, G = schema domain *Overvigilance and Inhibition*, H = quality of life, I = generalised anxiety symptoms, J = depressive symptoms. Letters represent the selected nodes and the lines between those nodes represent edges. Colours of the nodes represent the theoretical topics the nodes have been divided into. Green edges display positive partial correlations, whereas red edges display negative partial correlations. The thickness of the edges represents the indication of the strength of the edge. The thicker the edge, the higher the edge weight. The nodes with the highest degree centrality were schema domain *Impaired Autonomy and Performance* (D), schema domain *Overvigilance and Inhibition* (G), depressive symptoms (I), and schema domain *Disconnection and Rejection* (C). The nodes schema domains *Impaired Autonomy and Performance* (D), *Overvigilance and Inhibition* (G), and *Disconnection and Rejection* (C) have been found to have the highest strength index. The highest closeness and betweenness index were displayed by schema domains *Disconnection and Rejection* (C), *Impaired Autonomy and Performance* (D), *Overvigilance and Inhibition* (G). Based on the centrality indices, schema domains *Overvigilance and Inhibition* (G), *Impaired Autonomy and Performance* (D) and *Other-Directedness* (F) and experience of neglect (B) were identified as the most influential nodes. Bootstrap procedures were conducted for edge weight variation, the significance of edge weight and node strength differences, and the correlation stability of the centrality indices. The bootstrapping procedures indicated sufficient stability and interpretability of both the network and centrality indices, with the exception that the betweenness index shall not be interpreted.

whether there are meaningful differences between AFAB and AMAB transgender, as well as between transgender and cisgender individuals.

4.3. Schema domains are related to adverse childhood experiences and mental health conditions in adulthood

Further, to the best of our knowledge, this is the first study investigating the interplay between schema domains according to Young et al. (2003), adverse

childhood experiences and mental health conditions in adulthood using network analysis. Overall, results revealed a network with the schema domains *Overvigilance and Inhibition*, *Impaired Autonomy and Performance* and *Other-Directedness*, as well as the experience of neglect in childhood being identified as the most influential nodes among transgender individuals. The strongest connection has been observed between the adverse experience of abuse and neglect in childhood. Consistent with the literature, transgender individuals seem to more frequently experience higher childhood adversity (Schnarrs et al., 2019), as gender non-conformity seems to be an indicator of increased risk of childhood adversity (Roberts et al., 2012). The second strongest connection has been found between depressive and generalised anxiety symptoms. This is consistent with the current research, in which depression and anxiety display high comorbidity (Cramer et al., 2010) and serve as risk factors for each other (Jacobson & Newman, 2017). Considering adverse childhood experiences and mental health conditions in adulthood, the schema domain *Disconnection and Rejection* was related to experience of neglect in childhood and quality of life in adulthood, the schema domain *Impaired Autonomy and Performance* was related to experience of childhood abuse in childhood and depressive symptoms and quality of life in adulthood, as well as the schema domain *Overvigilance and Inhibition* was related to anxiety symptoms. The experience of childhood adversity seems to be positively related to the schema domains *Disconnection and Rejection* and *Impaired Autonomy and Performance*. This is in line with the systematic review by Pilkington, Bishop, et al., (2022), in which the experience of childhood adversity is linked to EMSs. Both schema domains are negatively related to the quality of life in adulthood. Further, Biedermann et al. (2021) reported that childhood adversities are associated with depression as well as suicidality in adulthood among transgender individuals. It is therefore likely that the experience of childhood adversity leads to the development of EMSs due to a lack of love, support, guidance, and belonging (schema domain *Disconnection and Rejection*), as well as a lack of self and self-agency (schema domain *Impaired Autonomy and Performance*), which interfere with mental health conditions and the quality of life in adulthood among transgender individuals. Surprisingly, the schema domain *Impaired Autonomy and Performance* was most strongly associated with depression. The current research suggest that depression is most strongly linked to EMSs belonging to the schema domain *Disconnection and Rejection* (Bishop et al., 2022), although Nicol et al. (2020) also found that depression was linked to EMSs belonging to the domain *Disconnection and Rejection* as well to the EMS *Failure*

belonging to the schema domain *Impaired Autonomy and Performance*. A possible explanation for this might be the positive correlation between both schema domains, implying as more schema domains interact with each other that the schema domains cannot be considered distinctively. Further, the schema domain *Overvigilance and Inhibition* was most strongly linked to anxiety. This implies that transgender individuals, who internalised strict rules and moral values as well as learn to suppress feelings and emotional expression, tend to develop more anxiety symptoms. Nicol et al. (2020) found that symptoms of anxiety were strongly associated with EMS of *Vulnerability to Harm/Illness* belonging to schema domain *Impaired Autonomy and Performance*. It can therefore be assumed that further research is necessary.

4.4. Limitations and recommendation for further research

The present study provides new and much-needed insights into the mental health and EMSs among transgender individuals. However, limitations must be considered. First, as a result of the cross-sectional design, no causality can be inferred. Even though network analysis displays significant connections between two nodes while controlling for other nodes in the network, it does not represent causal relationships. Further, as the study results are based on self-reporting, an objective verification is not possible. As all data presented were collected via an online survey, a selection bias cannot be ruled out. In addition, the *Essen Transidentity Quality of Life-Inventory* (Tagay et al., 2018) to investigate the transgender specific quality of life is currently only validated for AMAB transgender individuals, although a validation for AFAB transgender individuals is currently in working progress. Furthermore, the present study includes a heterogeneous sample of transgender individuals e.g. at different stages of transition, which needs to be considered when interpreting the results. Further, a limitation of the study is its focus on childhood maltreatment, which may not fully capture the specific experiences of transgender individuals. Emerging research highlights the potential impact of experiencing cisheterosexism and expressive suppression as an additional adversity (Charak et al., 2023; Schnarrs et al., 2022), which should be investigated further. Despite these limitations, this study provides especially new insight into the knowledge about the relationship between EMSs, adverse childhood experiences and mental health conditions in adulthood among transgender individuals. Further research is recommended and should consider the above-mentioned limitations. A study investigating the relationship of each EMS with adverse childhood experiences and mental health conditions in

adulthood among transgender individuals and in comparison, with matching cisgender individuals is recommended.

5. Conclusion

This is the first study investigating the relationship between EMSs, adverse childhood experiences and mental health conditions among transgender individuals using network analysis. In summary, this study highlights that transgender individuals are considered a vulnerable group and tend to have a high prevalence for childhood adversity as well as depressive and generalised anxiety symptoms, and low quality of life in adulthood. Transgender individuals seem to have an elevated pronounced level of EMSs, with AFAB transgender being in general more vulnerable than AMAB transgender. The network analysis revealed that especially the schema domains *Disconnection and Rejection* and *Impaired Autonomy and Performance* are related to experiencing childhood adversity and mental health conditions in adulthood. In conclusion, adverse childhood experiences emerged as having a central impact on the presentation of EMSs in adulthood, which interfere with mental health conditions in adulthood among transgender individuals. Regarding mental health care, adverse childhood experiences and EMSs should be taken into consideration in the treatment of current mental health conditions. The study findings have clinical implications that enable practitioners to tailor treatment to transgender individuals by considering specific factors as adverse childhood experiences and EMSs and recognising the unmet needs of transgender individuals, thereby allowing for more targeted interventions, e.g. within therapeutic relationships.

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Data availability statement

The participants of this study did not give written consent for their data to be shared publicly, so due to the sensitive nature of the research supporting data is not available.

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